

WATER IN THE ANTHROPOCENE

An Interdisciplinary Review of Availability, Quality, Health, Technology, and Governance

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Abstract

Water is simultaneously a molecular substance, a moving component of the Earth system, a foundation of public health and food production, an economic input, an ecological habitat, and a human right. This interdisciplinary review synthesizes evidence on global water distribution, hydrological change, water quality, water-related health, sectoral demand, treatment technologies, climate risk, and governance. The analysis is based on a structured narrative review of authoritative assessments and selected peer-reviewed literature, with emphasis on evidence available through June 2026. Although freshwater constitutes only about 2.5% of Earth's water, human systems depend disproportionately on a small and unevenly distributed portion of surface water, soil moisture, snow and ice, and accessible groundwater. Global progress in household water, sanitation and hygiene has been substantial but incomplete: in 2024, 2.1 billion people lacked safely managed drinking-water, 3.4 billion lacked safely managed sanitation, and 1.7 billion lacked basic hygiene services. Agriculture remains the dominant withdrawal sector, while pollution, groundwater depletion, cryosphere loss, urbanization, infrastructure deterioration, and climate-driven hydrological extremes intensify competition and uncertainty. Treatment advances—including membrane processes, advanced oxidation, decentralized systems, potable and non-potable reuse, and resource recovery—can expand reliable supply, but they cannot substitute for source protection, demand management, maintenance, affordability, and accountable institutions. The paper proposes an integrated water-security framework that links drivers, hydrological systems, service delivery, vulnerability, ecological integrity, and enabling governance. It concludes that durable water security requires a portfolio approach: protect catchments and environmental flows; reduce losses and inefficient demand; provide risk-based, fit-for-purpose treatment; scale safe reuse; strengthen monitoring and regulation; finance lifecycle maintenance; and prioritize populations facing structural disadvantage. Future research should better integrate blue water, green water, quality constraints, ecosystem thresholds, distributional justice, and decision-making under deep uncertainty.

Keywords: water; freshwater; water security; WASH; water quality; climate change; groundwater; wastewater reuse; desalination; governance

Highlights

- Freshwater security is constrained not only by physical scarcity, but also by contamination, infrastructure, affordability, institutions, and unequal exposure.
- The 2024 global household service gaps remain 2.1 billion for safely managed drinking-water, 3.4 billion for safely managed sanitation, and 1.7 billion for basic hygiene.
- Agriculture accounts for approximately 69% of global freshwater withdrawals, making food-system reform central to water security.
- Technology is most effective when embedded in source protection, risk-based regulation, lifecycle finance, and inclusive governance.

- An integrated framework is proposed to connect the water cycle, service systems, ecological limits, vulnerability, and corrective feedback.

Abbreviations

Table 1. Abbreviations used in the manuscript

Abbreviation	Meaning
AOP	Advanced oxidation process
AR6	Sixth Assessment Report of the Intergovernmental Panel on Climate Change
AQUASTAT	FAO Global Information System on Water and Agriculture
DALY	Disability-adjusted life year
E-flow	Environmental flow
IWRM	Integrated water resources management
JMP	WHO/UNICEF Joint Monitoring Programme
LCA	Life-cycle assessment
NF	Nanofiltration
RO	Reverse osmosis
SDG	Sustainable Development Goal
SEC	Specific energy consumption
UV	Ultraviolet
WASH	Water, sanitation and hygiene
WWDR	United Nations World Water Development Report

1. Introduction

Water connects the biosphere, atmosphere, cryosphere, lithosphere, and human economy. Its molecular polarity and hydrogen bonding produce an unusual combination of solvent capacity, heat storage, surface tension, phase behavior, and density variation that supports climate regulation and life. At societal scale, water is indispensable for drinking, sanitation, agriculture, energy, manufacturing, navigation, cultural practice, and ecosystem functioning. Yet the apparent abundance of water on a planet whose surface is about 71% water-covered conceals a central constraint: approximately 96.5% is held in the oceans, and only about 2.5% is freshwater. Most freshwater is stored in ice

or underground, leaving a very small fraction immediately accessible in rivers, lakes, wetlands, soils, and shallow aquifers [1].

The contemporary water challenge is therefore not reducible to a single global shortage. It is a set of interacting problems of timing, location, quality, access, reliability, affordability, governance, and ecological integrity. A basin may receive substantial annual precipitation yet experience seasonal scarcity, contamination, flood risk, or service exclusion. A city may possess adequate bulk supply but lose large volumes through leakage, intermittent distribution, illegal connections, or poor asset maintenance. A farming region may sustain production by mining groundwater, thereby transferring costs to future users and ecosystems. Water security is consequently a systems property rather than a volume alone.

Recent assessments show both progress and persistent inequality. Between 2000 and 2024, approximately 2.2 billion people gained safely managed drinking-water and 2.8 billion gained safely managed sanitation; nevertheless, in 2024, 2.1 billion people still lacked safely managed drinking-water, 3.4 billion lacked safely managed sanitation, and 1.7 billion lacked basic hygiene at home [5]. The deficits are concentrated in low-income countries, fragile contexts, rural areas, informal settlements, and populations facing discrimination or geographic isolation. At the same time, climate change is intensifying the hydrological cycle, altering precipitation, increasing the severity of some droughts and heavy rainfall events, accelerating glacier loss, and compounding risks to infrastructure, water quality, ecosystems, and health [2–4].

This paper provides a broad, publication-style review of water as a physical substance and a coupled natural-human system. It has four objectives: (i) synthesize the state of knowledge on water availability, quality, use, health, technology, and governance; (ii) distinguish physical scarcity from service and institutional scarcity; (iii) evaluate major intervention families and their trade-offs; and (iv) propose an integrated framework and research agenda suitable for interdisciplinary water-security analysis.

2. Scope and Review Method

The paper uses a structured narrative-review method rather than a formal systematic review or meta-analysis. Evidence was prioritized from authoritative international assessments, technical standards, and highly cited or recent peer-reviewed studies. Core institutional sources include the U.S. Geological Survey, WHO, UNICEF, UNESCO, UN-Water, FAO AQUASTAT, WMO, the IPCC, and United Nations human-rights resolutions. Peer-reviewed literature was selected to represent global scarcity, water-security threats, groundwater depletion, planetary freshwater change, water purification, desalination, and pollution-quality interactions [12–20, 26–29].

The evidence cutoff is 30 June 2026. Quantitative indicators are reported with their original reference year because monitoring cycles differ. No new field samples, surveys,

human-subject data, or laboratory experiments were produced. Figures 1–3 are original redrawings of publicly reported indicators; Figures 4–5 are conceptual syntheses. The review is global in scope and therefore cannot reproduce the hydrological, legal, cultural, and institutional detail required for a basin-specific management plan.

3. Water as a Physical and Earth-System Medium

3.1 Molecular properties and environmental significance

A water molecule consists of two hydrogen atoms covalently bonded to oxygen. The bent molecular geometry and unequal electron distribution create a dipole, enabling extensive hydrogen bonding. These intermolecular interactions explain much of water’s environmental importance: a high specific heat moderates climatic and physiological temperature fluctuations; high latent heats influence evaporation, condensation, storms, and energy transport; strong solvent capacity mobilizes nutrients and contaminants; surface tension affects capillarity and biological transport; and the lower density of ice relative to liquid water allows ice to float, insulating aquatic ecosystems during cold periods.

Table 2. Selected properties of water and their system-level implications

Property	Mechanism	Environmental or societal implication	Management relevance
Polarity and solvent capacity	Dipole interactions dissolve ions and polar compounds	Transports nutrients, salts, metals, organic pollutants and pathogens	Water-quality assessment must address mixtures and source-to-receptor pathways
High specific heat	Substantial energy is required to change temperature	Buffers climate and body temperature; stores thermal energy	Thermal pollution and warming affect oxygen, stratification and ecosystems
High latent heat	Large energy exchange during evaporation and condensation	Drives atmospheric circulation, rainfall and evaporative cooling	Evaporation losses matter in reservoirs, irrigation and cooling systems
Cohesion and adhesion	Hydrogen bonding and interaction with surfaces	Supports capillary rise, soil-water retention and plant transport	Soil structure and vegetation influence infiltration and green water

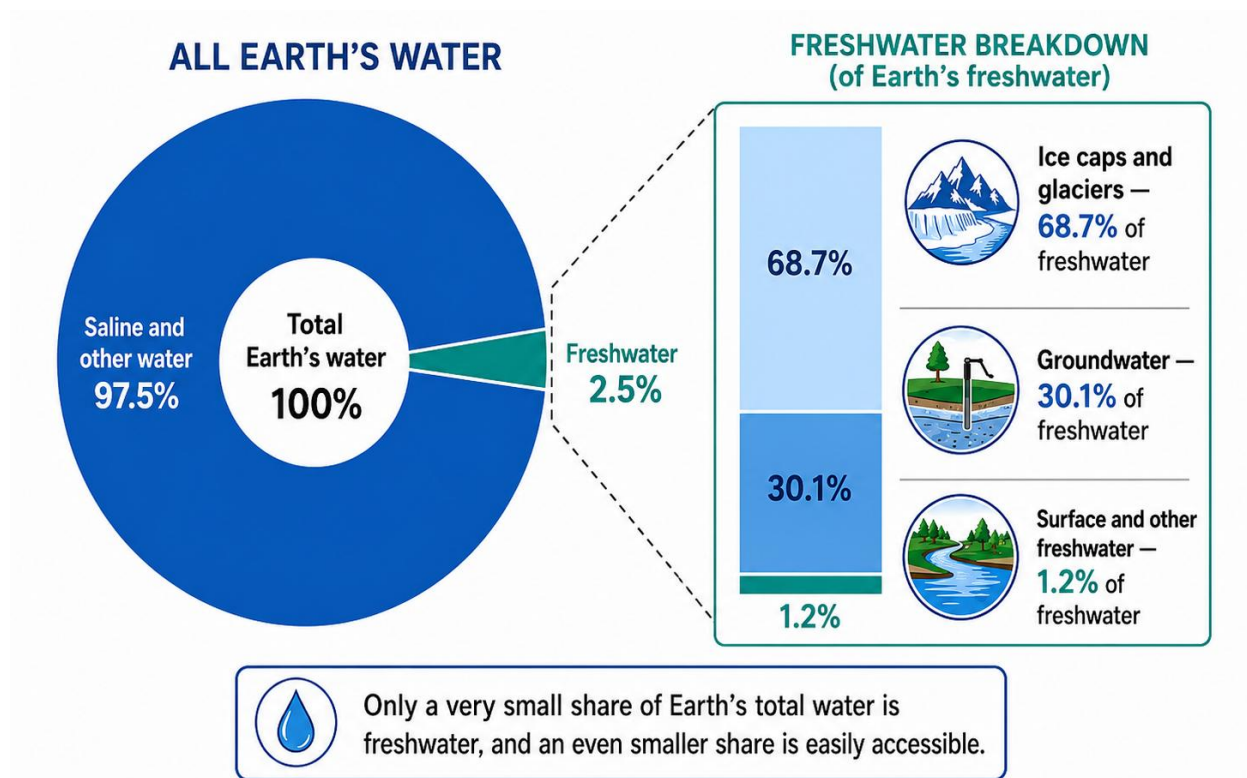
Property	Mechanism	Environmental or societal implication	Management relevance
Density anomaly	Ice is less dense than liquid water	Floating ice insulates lakes and supports seasonal turnover	Warming and ice-cover loss alter habitat and biogeochemistry
Low compressibility	Volume changes little under pressure	Efficient hydraulic transmission	Enables pressurized networks but contributes to water hammer risk

3.2 Distribution and the hydrological cycle

The global water cycle continuously exchanges water among ocean, atmosphere, land, groundwater, organisms, and ice. At the catchment scale, a simplified water-balance equation is:

$$dS/dt = P - ET - Q - W + R_{in} - R_{out} \quad (1)$$

where S is stored water, P is precipitation, ET is evapotranspiration, Q is runoff or streamflow, W is consumptive withdrawal, and R_{in} and R_{out} are subsurface or managed transfers into and out of the system. Equation (1) is deliberately general: real applications separate surface, soil, snow, reservoir, and groundwater stores and explicitly represent return flows, environmental requirements, and uncertainty.



Source: Rounded U.S. Geological Survey values.

Figure 1. The narrow fraction of Earth's water that is freshwater. Original graphic using rounded U.S. Geological Survey values [1].

The distinction between blue and green water is analytically useful. Blue water comprises liquid water in rivers, lakes, reservoirs, and aquifers that can be withdrawn or redirected. Green water is precipitation stored in soil and available to plants before returning through evapotranspiration. Food production and terrestrial ecosystems depend heavily on green water, while conventional planning has historically emphasized blue-water infrastructure. Recent planetary-boundary research argues that both streamflow and root-zone soil moisture must be considered because widespread departures from pre-industrial conditions can undermine Earth-system resilience [16,20].

4. The Contemporary State of Global Water Security

4.1 Scarcity, stress, and unequal exposure

Water scarcity is commonly divided into physical scarcity, where demand approaches or exceeds renewable supply, and economic or institutional scarcity, where water exists but infrastructure, finance, governance, or political inclusion prevents reliable access. The distinction is essential: household insecurity may coexist with water-intensive commercial uses, and high national availability can mask severe basin-level stress. Mekonnen and Hoekstra estimated that four billion people experience severe water scarcity for at least one month each year [12]. UN-Water reported that global

withdrawals represented 18.4% of renewable freshwater resources in 2018, but the aggregate conceals critical stress in Northern Africa and high stress in Central and Southern Asia [10].

A standard water-stress indicator is expressed as:

$$\text{Water stress (\%)} = \text{Freshwater withdrawals} / (\text{Renewable freshwater resources} - \text{Environmental flow requirements}) \times 100 \quad (2)$$

Indicator values should not be interpreted as a complete measure of security. Annual national ratios may conceal seasonal deficits, groundwater mining, transboundary dependence, water-quality constraints, ecosystem damage, or unequal allocation. Recent modeling also shows that pollution can transform nominal water availability into clean-water scarcity; one global study found that accounting for nitrogen pollution increased the number of water-scarce sub-basins from 984 to 2,517 for the 2010 baseline [17].

Table 3. Selected global water indicators and their interpretation

Indicator	Reported value	Reference year	Interpretive caution	Source
Earth's water in oceans	About 96.5%	Reference estimate	Does not represent usable freshwater	USGS [1]
Freshwater share of all water	About 2.5%	Reference estimate	Most is stored in ice or groundwater	USGS [1]
People lacking safely managed drinking-water	2.1 billion	2024	Service definition includes access on premises, availability and quality	WHO/UNICEF [5]
People lacking safely managed sanitation	3.4 billion	2024	Includes safe containment, emptying, transport and treatment requirements	WHO/UNICEF [5]
People lacking basic hygiene at home	1.7 billion	2024	Data completeness varies by country and income group	WHO/UNICEF [5]
Agricultural share of withdrawals	69%	Global AQUASTAT ratio	Regional shares differ sharply	FAO [8]
Global water stress	18.4%	2018	Global average masks subnational and seasonal extremes	UN-Water [10]

Indicator	Reported value	Reference year	Interpretive caution	Source
Freshwater flow supply originating in mountains	55–60%	WWDR 2025 synthesis	Depends on hydrological definition and spatial scale	UNESCO [4]
Countries reporting sufficient financial and human resources for WASH plans	Fewer than 13%	GLAAS 2025	Based on reporting countries and self-reported system assessments	WHO/UNICEF [7]

4.2 Groundwater, rivers, lakes, and the cryosphere

Groundwater stores approximately 30.1% of global freshwater and serves as a strategic buffer against rainfall variability [1]. It supplies households, irrigation, industry, and baseflow to rivers and wetlands. However, pumping can exceed recharge for decades because aquifer depletion is often invisible until wells deepen, land subsides, springs fail, or water quality deteriorates. Satellite gravimetry has revealed large regional trends in terrestrial water storage, with several drying hotspots associated with groundwater use and climate variability [27]. A 2024 global compilation of well observations found rapid groundwater declines to be widespread, particularly in dry agricultural regions, and reported that declines accelerated in 30% of aquifer systems with sufficient historical comparison [28].

Rivers and lakes provide only a small fraction of total freshwater storage but support a disproportionate share of direct human use and biodiversity. Dams, diversions, channelization, pollution, and land-cover change modify flow regimes, sediment transport, temperature, and connectivity. Earlier global analysis found that nearly 80% of the world’s population was exposed to high threat to human water security and that 65% of river discharge was under moderate to high biodiversity threat [13]. Technological investment can reduce service risk in wealthy regions without eliminating upstream ecological pressures, illustrating the difference between engineered security and basin health.

Mountains function as water towers by storing snow and ice and releasing meltwater seasonally. The 2025 WWDR estimates that mountains provide 55–60% of global annual freshwater-flow supply and that roughly two billion people depend on mountain waters [4]. Glacier retreat can temporarily increase meltwater before long-term decline, while reduced snowpack and earlier melt shift water away from periods of peak agricultural and urban demand. Risks include glacial-lake outburst floods, slope

instability, sediment changes, hydropower variability, and loss of culturally significant landscapes.

4.3 Freshwater change as an Earth-system concern

Water extraction is only one component of anthropogenic freshwater change. Land conversion, drainage, irrigation, dams, urban surfaces, atmospheric warming, and vegetation change alter streamflow and soil moisture. Porkka and colleagues estimated that annual local deviations in streamflow and soil moisture occurred across 18.2% and 15.8% of global land area, respectively, over the assessed recent period—well beyond the modeled pre-industrial variability envelope [16]. Such changes affect habitat, carbon cycling, rainfall recycling, fire risk, and climate regulation. The emerging implication is that water management should protect characteristic ranges and timing of flow and soil moisture, not merely annual withdrawal volumes.

5. Water Quality, Pollution, and Ecosystem Health

5.1 From source contamination to exposure

Water quality links catchment activity to human and ecological exposure. Contaminants may originate from geology, agriculture, sanitation failures, mining, industry, waste disposal, urban runoff, atmospheric deposition, or treatment and distribution systems. Their significance depends on concentration, persistence, transformation products, co-exposure, pathogen viability, and the vulnerability of receptors. A risk-based approach therefore begins with system assessment from catchment to consumer, defines health-based targets, implements control measures, and uses independent surveillance—the framework reinforced in WHO’s 2026 drinking-water guidelines [6].

Table 4. Major water-contaminant classes, sources, and management responses

Class	Examples	Common sources	Primary concerns	Priority controls
Microbial	Bacteria, viruses, protozoa, helminths	Human and animal excreta, sewer overflows, inadequate treatment	Acute infection, outbreaks, child mortality	Sanitation, source protection, filtration, disinfection, residual control

Class	Examples	Common sources	Primary concerns	Priority controls
Nutrients	Nitrate, ammonia, phosphate	Fertilizer, manure, wastewater, aquaculture	Eutrophication, hypoxia, harmful algal blooms, infant methemoglobinemia	Nutrient management, wastewater treatment, buffers, wetland restoration
Geogenic chemicals	Arsenic, fluoride, iron, manganese, salinity	Aquifer minerals, geothermal sources, seawater intrusion	Chronic toxicity, dental/skeletal effects, acceptability problems	Alternative source, selective removal, blending, managed pumping
Metals and metalloids	Lead, mercury, cadmium, chromium	Mining, industry, corrosion, waste	Neurological, renal, developmental and ecological effects	Pollution prevention, corrosion control, precipitation/adsorption, monitoring
Organic micropollutants	Pesticides, pharmaceuticals, solvents, PFAS and other persistent compounds	Agriculture, industry, consumer products, landfills, wastewater	Chronic and mixture effects; persistence and bioaccumulation	Source restrictions, activated carbon, membranes, AOPs, producer responsibility
Physical pollutants	Sediment, heat, plastics and microplastics	Erosion, construction, thermal discharges, waste fragmentation	Habitat degradation, transport of sorbed pollutants, ecological stress	Erosion control, thermal management, waste prevention, capture and monitoring
Salinity	Dissolved salts	Irrigation return flow, mining, road salts, seawater intrusion	Crop loss, infrastructure corrosion, ecosystem shifts	Drainage management, source substitution, desalination, aquifer protection

5.2 Wastewater, ambient quality, and reuse

Untreated or inadequately treated wastewater transfers pathogens, oxygen-demanding material, nutrients, pharmaceuticals, metals, and other pollutants into rivers, lakes, coasts, and groundwater. UN-Water's 2024 update emphasized large data gaps: among 22 countries reporting industrial wastewater treatment, 38% was reported as treated and 27% as safely treated [9]. Monitoring of ambient water quality improved to 120 reporting countries in the 2023 data drive, but many low- and middle-income countries still lack sufficient stations, laboratory capacity, data systems, and long-term financing [11].

Wastewater is also a resource containing water, nutrients, organic carbon, and heat. Reuse can reduce freshwater withdrawals and pollution when treatment is matched to exposure pathways and backed by monitoring, operational competence, and public trust. Fit-for-purpose reuse distinguishes irrigation, industrial, environmental, and potable applications rather than applying one quality standard to all uses. Important safeguards include source control, multiple treatment barriers, validated pathogen reduction, salinity management, online monitoring, contingency plans, and clear institutional responsibility.

5.3 Freshwater ecosystems and environmental flows

Healthy rivers, floodplains, wetlands, lakes, riparian zones, and aquifers perform water-supply functions that engineered systems often attempt to replace: filtration, nutrient retention, flood attenuation, groundwater recharge, habitat provision, shoreline stabilization, and temperature moderation. Environmental flows are the quantity, timing, and quality of water required to sustain freshwater and estuarine ecosystems and the livelihoods that depend on them. Allocations that treat environmental water as a residual after consumptive demand are prone to ecological collapse and long-term service loss. Environmental-flow requirements should therefore be embedded in water accounts and stress indicators, as in Equation (2), and supported by enforceable operating rules.

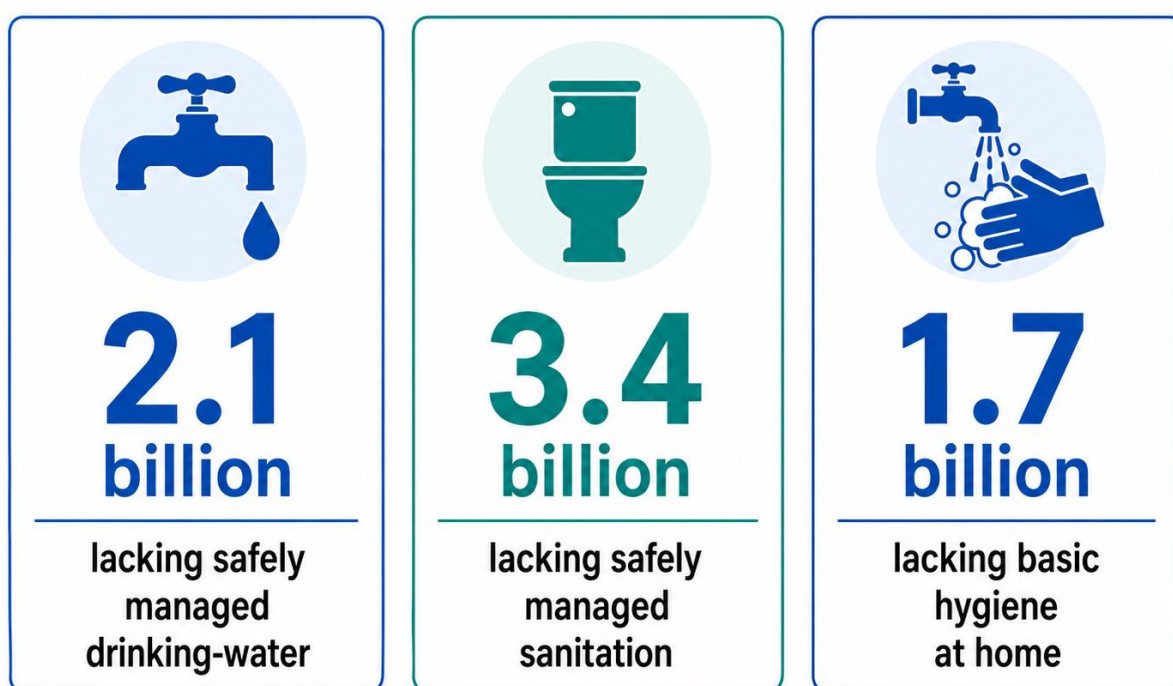
6. Water, Health, Equity, and Human Rights

6.1 Health pathways

Unsafe water and inadequate sanitation and hygiene contribute to diarrhoeal disease, cholera, typhoid, hepatitis, parasitic infections, neglected tropical diseases, undernutrition, and health-care-associated infection. Chemical hazards such as arsenic, fluoride, nitrate, and lead can cause chronic effects that are less visible than outbreaks but equally consequential. WHO reports that inadequate WASH continues to account for more than one million diarrhoeal deaths each year [31]. Health burdens also arise indirectly through time spent collecting water, school absence, physical injury, violence

risk, heat exposure, and reduced capacity to manage menstruation or care for sick family members.

A service-level perspective is more informative than a binary improved/unimproved distinction. Safely managed drinking-water requires an improved source that is accessible on premises, available when needed, and free from fecal and priority chemical contamination. Safely managed sanitation requires hygienic separation of excreta and safe management through containment, emptying, transport, treatment, and disposal or reuse. These definitions shift attention from infrastructure installation to continuous performance.



WASH = water, sanitation, and hygiene.

Source: WHO/UNICEF JMP estimates, 2024.

Figure 2. Global household WASH service gaps in 2024. Original graphic based on WHO/UNICEF JMP estimates [5].

6.2 Inequality and the human-rights framework

The United Nations General Assembly recognized safe and clean drinking water and sanitation as a human right essential to life and all human rights in Resolution 64/292 (2010) [22]. Later resolutions articulated distinct rights to safe drinking water and sanitation and emphasized sufficient, safe, acceptable, physically accessible, and affordable services without discrimination. A rights-based approach requires more than aggregate coverage: it demands attention to participation, accountability, transparency,

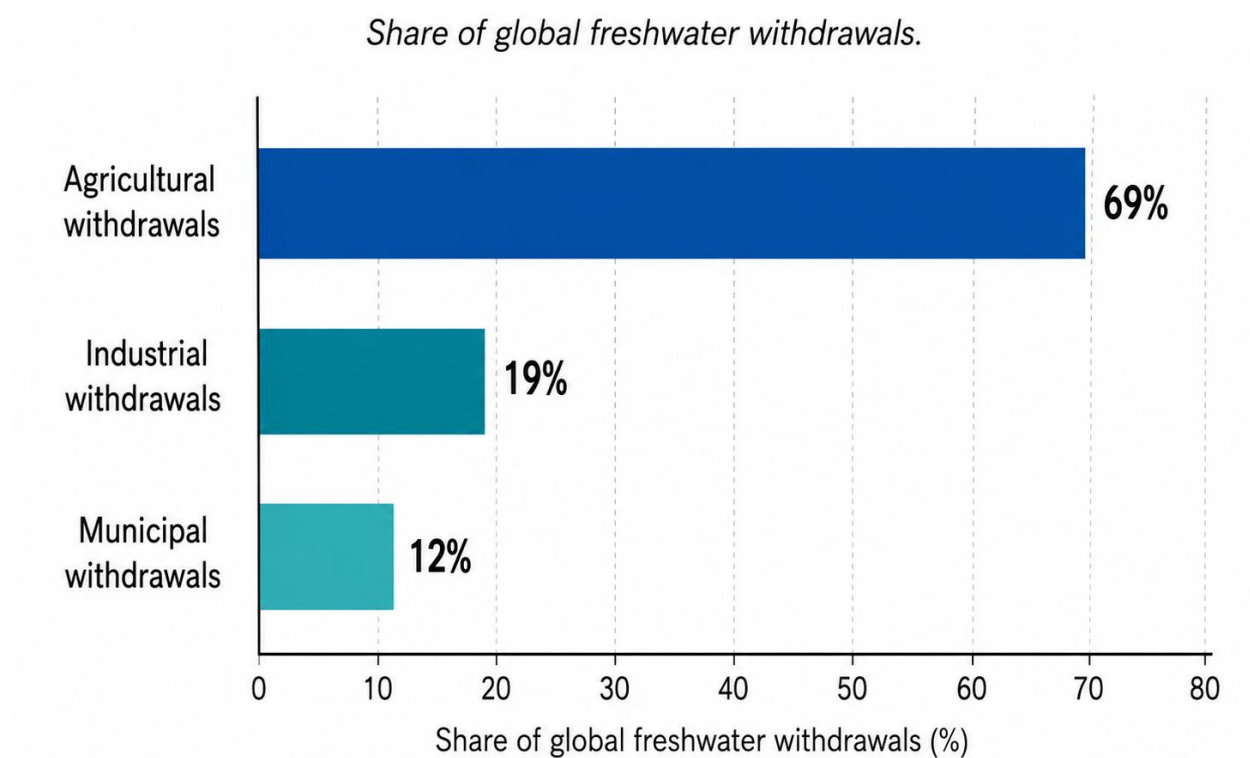
non-discrimination, affordability, privacy, dignity, and progressive realization using the maximum available resources.

Inequality appears at multiple scales. Rural households may face long collection times and unreliable boreholes; urban informal settlements may pay more per liter to vendors than networked households; Indigenous and local communities may lose customary access through dams, mining, or land conversion; and women and girls often bear disproportionate collection and care burdens. Tariff reform must therefore balance cost recovery with affordability. Lifeline quantities, targeted subsidies, cross-subsidies, social protection, and transparent connection policies can protect low-income users more effectively than blanket underpricing that starves utilities of maintenance finance.

7. Water in Food, Energy, Industry, and Cities

7.1 Agriculture and food systems

Agriculture accounts for approximately 69% of global freshwater withdrawals, with substantially higher shares in parts of Africa and Asia [8]. Irrigation stabilizes yields and enables high-value production, but inefficient conveyance, poorly designed subsidies, inappropriate crop choices, salinization, waterlogging, and unregulated pumping can undermine basin sustainability. Farm-level efficiency does not automatically create basin-level savings: water ‘lost’ from a field may return to rivers or aquifers and be reused downstream. Real water savings require reductions in non-beneficial consumption or withdrawals that would otherwise be unavailable for environmental flows or other users.



Source: FAO AQUASTAT global ratios.

Figure 3. Global freshwater withdrawal by major sector. Original graphic based on FAO AQUASTAT global ratios [8].

Priority strategies include soil-moisture conservation, precision irrigation, deficit irrigation where agronomically appropriate, scheduling using weather and crop data, leakage control in canals, conjunctive use of surface water and groundwater, salinity management, drought-tolerant varieties, adjusted planting calendars, and incentives aligned with basin limits. Dietary patterns, food loss, trade, and procurement policies also shape water demand through virtual-water flows. Policy should avoid shifting pressure from one basin or population to another without transparent accounting.

7.2 Energy and industry

Water and energy systems are interdependent. Water is used for fuel extraction, refining, cooling, hydropower, bioenergy, and manufacturing, while energy is required for pumping, conveyance, treatment, desalination, heating, and wastewater management. Thermal power plants are sensitive to low flows and high intake temperatures; hydropower depends on runoff and reservoir operations; and desalination can increase electricity demand. Integrated planning should evaluate consumptive use, return-flow temperature and quality, carbon intensity, and reliability under drought and flood scenarios.

Industrial water management follows a hierarchy: prevent contamination and unnecessary demand; segregate clean and contaminated streams; optimize processes and cooling; reuse internally; recover materials and heat; treat residuals; and disclose withdrawals, consumption, discharge, and basin context. Corporate water targets based only on facility efficiency can be misleading if operations remain located in stressed basins or supply chains transfer water risk to farmers and communities.

7.3 Urban water systems

Urban water security depends on source diversification, reliable treatment, distribution integrity, drainage, wastewater management, and land-use planning. Rapid growth often outpaces networks, producing intermittent supply, illegal or unsafe connections, sewer overflows, and dependence on tankers or private wells. Intermittency can create negative pressure that draws contaminants into pipes. The 2025 GLAAS assessment reported average non-revenue water of 39% among reporting countries and identified major financing and regulatory gaps [7]. Reducing physical leakage and improving metering, billing, pressure management, and repair response can provide water more rapidly and cheaply than developing a new source in many systems.

Water-sensitive urban design combines conventional ‘gray’ infrastructure with green and blue systems: permeable surfaces, rain gardens, urban wetlands, detention spaces, restored streams, rainwater harvesting, managed aquifer recharge, and flood-compatible public space. These measures can reduce peak runoff and heat while supporting amenity and biodiversity, but they require maintenance, land governance, sediment and mosquito management, and design for extreme events.

8. Water Treatment, Supply Augmentation, and Resource Recovery

8.1 Treatment as a multiple-barrier system

No single treatment process removes all hazards. Reliable drinking-water systems use multiple barriers: catchment protection; appropriate abstraction; coagulation and clarification where needed; filtration; disinfection; corrosion control; secure storage; distribution residuals or equivalent controls; and operational monitoring. Groundwater may require aeration, adsorption, ion exchange, softening, or membrane treatment depending on geochemistry. Treatment selection should be based on raw-water variability, target hazards, scale, operator capacity, supply chains, energy, residuals, and lifecycle cost rather than nominal removal performance alone.

$$\text{Removal efficiency (\%)} = (C_{\text{in}} - C_{\text{out}}) / C_{\text{in}} \times 100 \quad (3)$$

Equation (3) is useful for process evaluation but insufficient for health assurance when influent concentration varies or detection limits are high. For pathogens, log-reduction

values and quantitative microbial risk assessment are often more appropriate. For chemical mixtures, treatment by-products and transformation products must also be considered.

8.2 Membranes, desalination, and advanced processes

Membrane technologies span microfiltration, ultrafiltration, nanofiltration, and reverse osmosis. They can provide compact, high-quality treatment and are central to desalination and advanced reuse, but performance is constrained by fouling, scaling, energy, concentrate management, membrane integrity, and skilled operation. Foundational reviews identified the need for purification with lower energy, fewer chemicals, and safer reuse [14]. Reverse-osmosis desalination has improved substantially, yet thermodynamic limits, pretreatment, intake impacts, carbon intensity, and brine disposal remain important [15].

$$\text{Specific energy consumption, } SEC = \text{Electrical energy input} / \text{Product-water volume} \quad (4)$$

Desalination is most defensible where conservation, leakage reduction, reuse, and lower-impact sources cannot meet reliability objectives and where energy and brine impacts are controlled. Renewable electricity can reduce operational emissions but does not eliminate marine intake, chemical use, or concentrate concerns. Brine may contain recoverable materials, yet resource recovery should be evaluated through mass balance, market demand, toxicity, and full lifecycle assessment rather than assumed to be beneficial.

8.3 Decentralized treatment and household systems

Decentralized systems can serve rural settlements, peri-urban areas, emergency contexts, institutions, or buildings that are costly to connect. Options include protected springs, boreholes, small piped schemes, chlorination, ceramic or membrane filters, UV, solar disinfection, constructed wetlands, packaged treatment, and fecal-sludge management. Their main failure modes are often institutional rather than technical: absent operators, inconsistent consumables, broken pumps, weak spare-parts markets, poor sludge management, and unclear ownership. Service delivery models should finance maintenance, test water quality, disclose performance, and define response responsibilities.

8.4 Wastewater reuse and circularity

Reuse portfolios may include industrial cooling, landscape irrigation, agriculture, groundwater recharge, environmental enhancement, indirect potable reuse, and direct

potable reuse. Potable reuse typically uses advanced treatment such as microfiltration or ultrafiltration, reverse osmosis or an alternative robust barrier, advanced oxidation, disinfection, storage or environmental buffers where applicable, and continuous monitoring. Public acceptance is strengthened by transparent performance data, independent oversight, meaningful engagement, and clear comparison with existing source-water risks.

Table 5. Comparison of major treatment and supply options

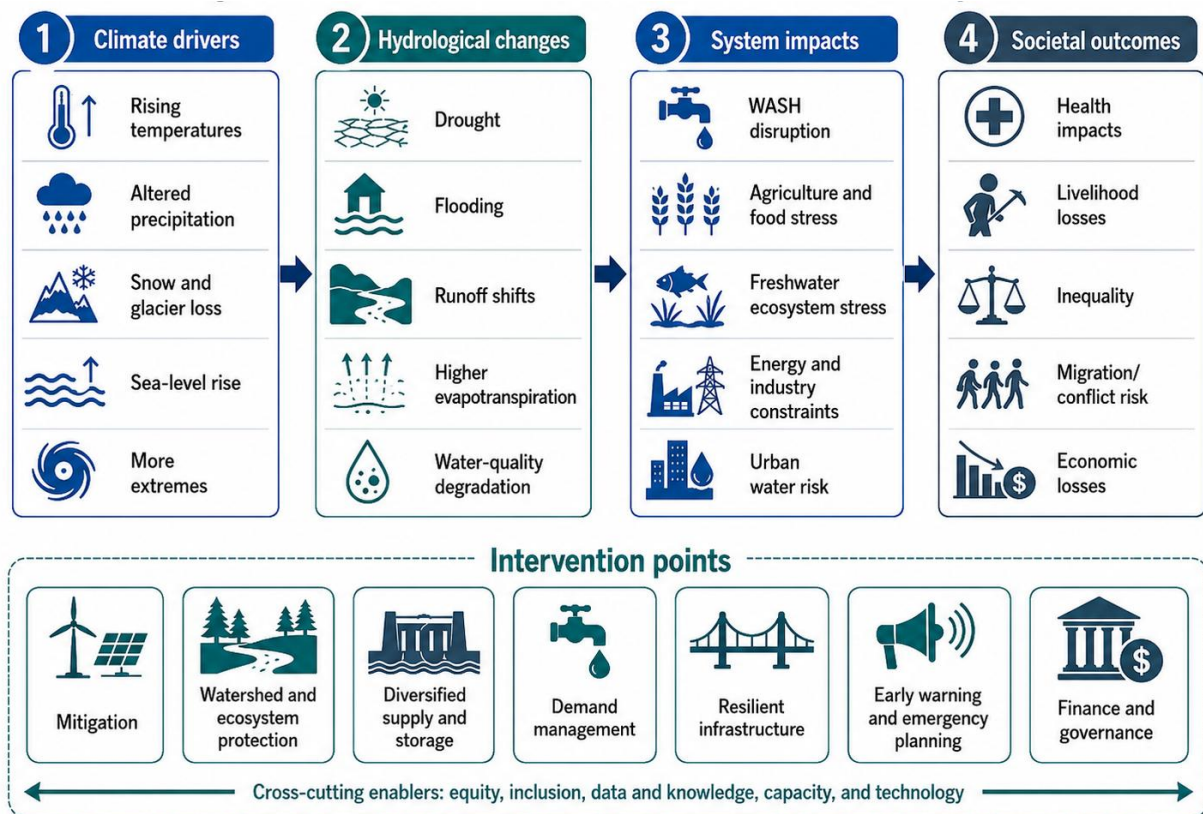
Option	Best suited to	Strengths	Key limitations	Critical safeguards
Conventional coagulation–filtration–disinfection	Surface water with turbidity and microbial risk	Mature, scalable, robust when well operated	Sludge, chemical supply, variable micropollutant removal	Jar testing, filter monitoring, disinfectant control, residual management
Activated carbon / adsorption	Taste, odor, natural organic matter and many organics	Flexible polishing; granular media can be regenerated	Media exhaustion; selective performance	Breakthrough monitoring, source control, media replacement
Membrane filtration (MF/UF/NF)	Particles, pathogens, hardness and selected dissolved contaminants	Compact and high-quality effluent	Fouling, concentrate, integrity and energy	Pretreatment, integrity tests, cleaning and residual plan
Reverse osmosis	Desalination and advanced reuse	Broad dissolved-solute rejection	High pressure, concentrate, boron and small neutral compounds may need attention	Energy recovery, pretreatment, remineralization, brine management
Advanced oxidation	Trace organics and oxidation-resistant compounds	Can destroy rather than transfer contaminants	Energy, by-products and incomplete mineralization	Dose validation, by-product monitoring, downstream biofiltration where appropriate

Option	Best suited to	Strengths	Key limitations	Critical safeguards
Constructed wetlands / nature-based treatment	Wastewater polishing, runoff and decentralized contexts	Low energy; habitat and co-benefits	Land, seasonality, hydraulic short-circuiting, maintenance	Pretreatment, loading control, vector and vegetation management
Managed aquifer recharge	Seasonal storage, drought buffers and water-quality improvement	Reduces evaporation; can stabilize groundwater	Clogging, geochemical reactions, contamination and property rights	Source quality, hydrogeology, monitoring wells, recovery rules
Demand management and leakage control	All systems, especially constrained networks	Often least-cost and rapid; avoids new withdrawals	Requires data, metering, institutions and behavior change	Equity protections, pressure management, verified savings

9. Climate Change, Hydrological Extremes, and Resilience

Climate change is intensifying the water cycle, with more intense heavy precipitation and associated flooding in many regions, more severe drought in some regions, altered monsoons, snow and glacier loss, and changes in soil moisture and evapotranspiration [2]. Impacts depend on warming level, local climate, land management, infrastructure, institutions, and exposure. Higher temperatures can increase water demand and evaporation, reduce dissolved oxygen, stimulate harmful algal blooms, accelerate some chemical reactions, and challenge treatment. Floods can overwhelm sanitation, mobilize sediments and hazardous materials, and damage wells and networks. Drought can concentrate pollutants, reduce dilution, deepen groundwater pumping, and increase wildfire-related contamination.

IPCC assessments conclude that climate change will exacerbate existing urban water-service challenges and increase risks to freshwater ecosystems, WASH services, energy production, health, migration, and conflict in context-dependent ways [3]. Recent global modeling estimated a baseline water gap of 457.9 km³/year, rising by 5.8% under 1.5°C warming and 14.7% under 3°C warming, with large regional differences and uncertainty [18]. The WMO reported that only about one-third of global river basins experienced normal conditions in 2024, continuing a multiyear pattern of imbalance [30].



Original synthesis informed by IPCC, UNESCO, and WMO assessments.

Figure 4. Climate–water risk cascade and intervention points. Original synthesis informed by IPCC, UNESCO and WMO assessments [2–4,30].

Resilience should not be equated with simply building larger infrastructure. A robust portfolio combines mitigation of greenhouse-gas emissions; watershed protection; environmental flows; diversified sources; aquifer storage; demand management; modular treatment; flood-safe design; backup power and chemicals; asset redundancy; emergency interconnections; drought and flood triggers; early warning; insurance or contingency finance; and plans for vulnerable populations. Decisions should be tested across multiple plausible futures rather than optimized for a single forecast.

10. Governance, Economics, and Institutions

10.1 Allocation and integrated management

Water allocation establishes who may use how much water, when, where, for what purpose, and under what conditions. Effective regimes define entitlements, measurement requirements, environmental reserves, drought-sharing rules, transfer mechanisms, and enforcement. Informal and customary rights must be recognized where they are legitimate and socially embedded. Integrated water resources management seeks coordination across sectors and scales, but integration can become procedural rather than operational unless authority, budgets, data, and accountability are aligned.

Basin institutions face collective-action problems because upstream actions affect downstream users, surface water and groundwater are connected, and political boundaries rarely follow hydrology. Transboundary cooperation benefits from joint fact-finding, shared monitoring, notification procedures, benefit sharing, data protocols, dispute resolution, and flexible rules for extremes. Water diplomacy is more durable when it distributes benefits and risks rather than focusing only on volumetric division.

10.2 Finance, regulation, and lifecycle performance

Capital investment receives more political attention than operation and maintenance, yet service failure usually emerges from neglected assets, insufficient operators, weak procurement, power interruptions, and absent monitoring. Lifecycle finance should cover source protection, treatment, networks, sanitation chains, laboratories, information systems, sludge and concentrate management, renewal, and emergency response. The 2025 GLAAS findings—fewer than 13% of countries reporting sufficient financial and human resources for WASH plans—illustrate the implementation gap between policy and delivery [7].

Regulation should be risk-based, transparent, independent enough to manage conflicts of interest, and capable of escalating enforcement. Water-quality standards alone are insufficient without surveillance, laboratory quality assurance, incident reporting, public disclosure, and corrective action. Economic instruments—including abstraction charges, pollution fees, tradable entitlements, block tariffs, connection subsidies, and payments for watershed services—can support objectives but require careful design to prevent exclusion, speculation, or the commodification of essential household needs.

10.3 Data, digital systems, and accountability

Water decisions depend on observations of precipitation, streamflow, groundwater levels, withdrawals, quality, ecosystems, assets, pressure, consumption, and service interruptions. Remote sensing, smart meters, supervisory control systems, low-cost sensors, machine learning, and digital twins can improve situational awareness. However, digital systems can reproduce bias, exclude unconnected populations, create cybersecurity risks, and generate false precision. Data governance should specify validation, interoperability, privacy, ownership, public access, and the decisions that measurements are intended to support.

11. An Integrated Framework for Water Security

The paper proposes that water security be evaluated as the capacity of a coupled natural-human system to provide sufficient, safe, accessible, reliable, and affordable water and

sanitation services while maintaining ecosystem functions and managing water-related hazards. A compact conceptual expression is:

$$WS = f(Q_n, Q_l, A, R, F, E, G; V) \quad (5)$$

where WS is water security; Q_n is quantity; Q_l is quality; A is accessibility; R is reliability and resilience; F is affordability; E is ecosystem integrity; G is governance capacity; and V represents vulnerability and distributional conditions. Equation (5) is not a predictive formula; it is a checklist against one-dimensional assessments. High performance in one dimension cannot fully compensate for failure in another. For example, abundant but contaminated water is insecure, affordable but intermittent water may be unsafe, and reliable supply that destroys downstream ecosystems is not sustainable.

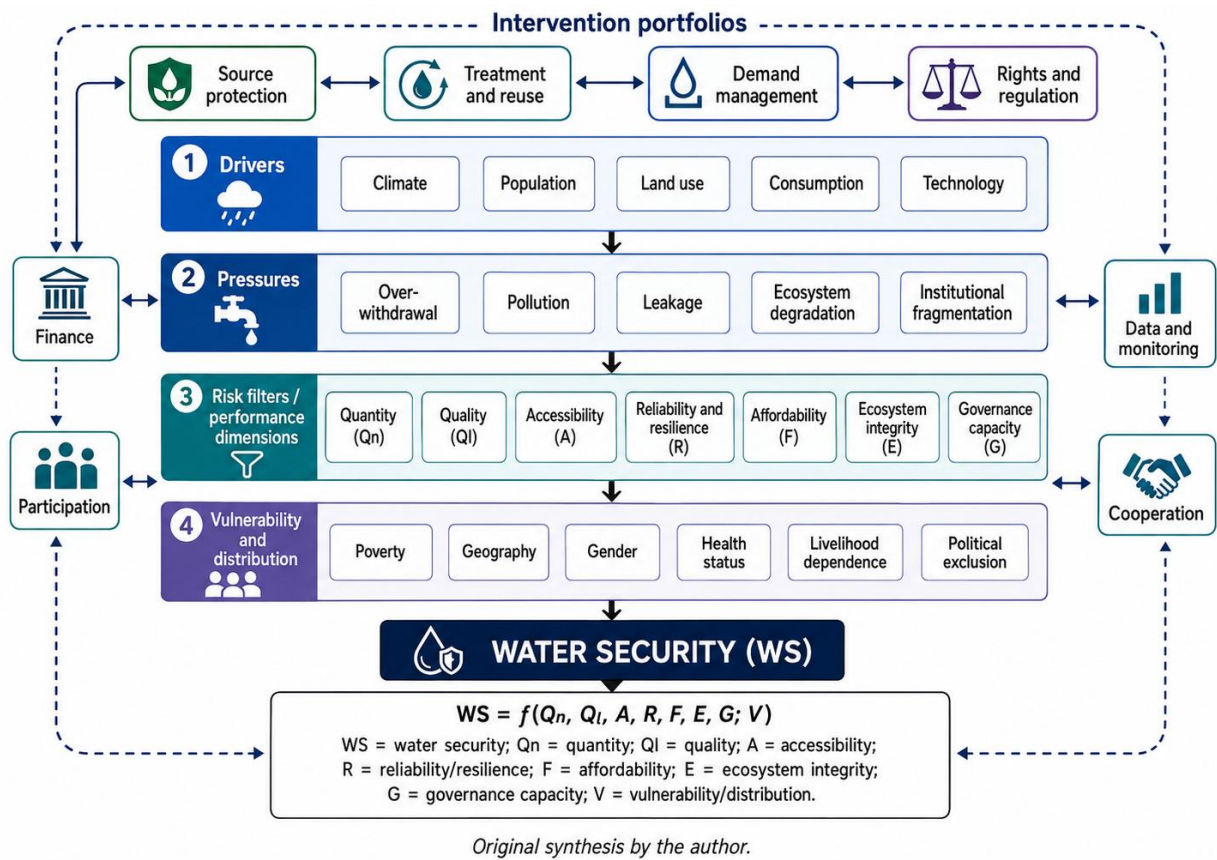


Figure 5. Integrated water-security framework. Original synthesis by the author.

The framework separates four analytical layers. First, exogenous and endogenous drivers—climate, population, land use, consumption and technology—modify the water system. Second, pressures such as over-withdrawal, pollution, leakage and fragmentation affect hydrological and service performance. Third, risk filters determine whether water is sufficient, safe, accessible, reliable and affordable. Fourth, vulnerability shapes who experiences harm. Intervention portfolios create corrective feedback through source protection, service innovation, and enabling systems of rights, regulation, finance, data, participation, and cooperation.

12. Strategic Priorities

Table 6. Priority actions for a water-secure transition

Priority	Near-term action	Medium-term institutional change	Performance indicators
Protect sources and ecosystems	Map critical catchments, pollution hotspots, recharge zones and environmental-flow deficits	Embed enforceable source-protection and environmental-flow rules in allocation and land-use planning	Water-quality trends; e-flow compliance; wetland and riparian condition
Close basic WASH gaps	Target unserved populations; restore failed systems; strengthen sanitation and hygiene services	Professionalize rural and small-system service models; establish affordability mechanisms	Safely managed service levels; continuity; water quality; collection time; affordability
Reduce avoidable demand and loss	Repair priority leaks; pressure management; irrigation scheduling; industrial audits	Align tariffs, subsidies and abstraction rules with verified basin savings and equity	Non-revenue water; consumptive use; crop water productivity; verified savings
Scale safe reuse	Identify fit-for-purpose reuse opportunities and high-value early projects	Adopt health-based regulations, operator certification, monitoring and transparent risk communication	Reuse volume; compliance; energy; residuals; public confidence
Manage groundwater conjunctively	Meter high-volume abstraction; protect recharge; drought triggers	Link groundwater and surface-water rights; managed aquifer recharge and land-use controls	Water-level trends; abstraction/recharge balance; land subsidence; salinity
Build climate resilience	Stress-test assets; emergency plans; backup power and chemicals; early warning	Integrate water into climate finance, urban planning and disaster risk management	Outage duration; people protected; forecast lead time; recovery time

Priority	Near-term action	Medium-term institutional change	Performance indicators
Strengthen institutions and finance	Publish system performance and funding gaps; prioritize maintenance	Independent regulation, lifecycle budgeting, workforce development and data governance	Budget execution; staffing; compliance reporting; asset condition; complaint resolution

Implementation should follow a hierarchy. Prevent degradation before treating it; manage demand before developing high-impact supply; match quality to use; recover and reuse where safe; use natural storage and ecosystems where appropriate; and build major infrastructure only after alternatives and induced-demand risks are assessed. The hierarchy is not absolute—acute shortages may require parallel action—but it improves the probability that investments address root causes.

13. Research Agenda

Despite rapid advances, major knowledge and implementation gaps remain. Priority research areas include:

- Integrated blue–green water accounting that includes quality constraints, environmental flows, return flows, and uncertainty at decision-relevant spatial and temporal scales.
- Causal evidence on which combinations of tariffs, subsidies, regulation, information, and technology produce real basin-level water savings without harming low-income households or food security.
- Low-energy and low-chemical treatment for persistent and mobile contaminants, with reliable performance in small and resource-constrained systems.
- Monitoring and health evidence for complex mixtures, antimicrobial resistance, microplastics, algal toxins, and treatment transformation products.
- Operationally useful climate information, including compound drought–heat–fire–flood events, non-stationary design standards, and adaptation limits.
- Groundwater governance and recovery pathways, including managed recharge, demand reduction, land subsidence, salinity, and ecosystem dependence.
- Distributional and procedural justice metrics that reveal who benefits, who pays, who participates, and whose cultural and ecological values are recognized.
- Lifecycle and circular-economy assessment of desalination, reuse, nutrient recovery, brine management, and nature-based solutions.

- Open, interoperable water data systems with uncertainty communication, privacy protection, cybersecurity, and community-based observation.
- Comparative studies of institutional capacity, maintenance models, workforce development, and regulatory independence across service scales.

14. Limitations

This review is intentionally broad. It does not apply a registered systematic-review protocol, quantify publication bias, or conduct a meta-analysis. Global indicators combine national datasets with uneven monitoring capacity and may lag current local conditions. Water terminology and metrics differ across organizations and disciplines. The treatment comparison is qualitative and cannot replace site-specific pilot testing, risk assessment, hydrogeological investigation, economic analysis, environmental assessment, or consultation. Climate projections and future water-gap estimates contain scenario, model, socioeconomic, and downscaling uncertainty. Finally, the integrated framework is a conceptual synthesis requiring empirical testing across basins, utilities, and governance settings.

15. Conclusion

Water is abundant in planetary terms but scarce in the forms, places, times, and qualities required by people and ecosystems. The global challenge is not merely to find more water; it is to govern a dynamic cycle under intensifying human pressure and climate change. Household WASH gains demonstrate that progress is possible, yet billions remain without safely managed services. Meanwhile, agricultural demand, groundwater depletion, pollution, ecological alteration, cryosphere loss, urban growth, and weak maintenance threaten the reliability of future supply.

The evidence supports a portfolio rather than a single technological solution. Source protection and environmental flows preserve the natural system. Demand management and leakage reduction reduce avoidable pressure. Risk-based treatment, reuse, desalination, storage, and digital monitoring can increase reliability when matched to context and lifecycle capacity. Rights, affordability, participation, regulation, finance, workforce, and transparent data determine whether infrastructure becomes a durable service. Water security should therefore be judged by quantity, quality, accessibility, reliability, affordability, ecosystem integrity, governance, and distributional justice together.

The central policy principle is simple: protect water before it is polluted, use it more productively before expanding withdrawals, treat it according to risk and intended use, return it safely to the cycle, and ensure that the least powerful are not left with the highest costs and risks. Achieving this principle requires sustained institutions and

knowledge, not only projects. Water management in the Anthropocene must be adaptive, precautionary, scientifically grounded, and explicitly concerned with both human dignity and the living systems on which all water services depend.

Declarations

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Consent to participate and publish: Not applicable.

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Appendix A. Data Used in Original Quantitative Figures

Table A1. Figure source values

Figure	Measure	Value	Unit	Source
Figure 1	Saline and other water	97.5	% of all water	USGS [1]
Figure 1	Freshwater	2.5	% of all water	USGS [1]
Figure 1	Ice caps and glaciers	68.7	% of freshwater	USGS [1]
Figure 1	Groundwater	30.1	% of freshwater	USGS [1]
Figure 1	Surface and other freshwater	1.2	% of freshwater	USGS [1]
Figure 3	Agricultural withdrawals	69	% of global withdrawals	FAO AQUASTAT [8]
Figure 3	Industrial withdrawals	19	% of global withdrawals	FAO AQUASTAT [8]
Figure 3	Municipal withdrawals	12	% of global withdrawals	FAO AQUASTAT [8]
Figure 2	Lacking safely managed drinking-water	2.1	billion people, 2024	WHO/UNICEF [5]

Figure	Measure	Value	Unit	Source
Figure 2	Lacking safely managed sanitation	3.4	billion people, 2024	WHO/UNICEF [5]
Figure 2	Lacking basic hygiene at home	1.7	billion people, 2024	WHO/UNICEF [5]